1. Purpose and Scope

This document explores Collatz dynamics through a reduced structural formalism that factors out deterministic valuation behavior (“extended valuation blocks”) and studies the induced dynamics on a reduced state (ω,ℓ). The aim is to expose arithmetic structure hidden by step-by-step iteration.

No claims are made here regarding convergence or resolution of the Collatz conjecture; the focus is solely on exposing arithmetic structure in the reduced dynamics.

2. Layered View of Collatz Map (Important)

We explore the dynamic of Collatz at 3 levels of abstraction as follows:

1) **Collatz Map iterations** $T(x) = x^{+}$ are as presented in the original problem, the fundamental operation layer,
2) **Extended Valuation blocks** follow the basic observation that any odd operation is immediately followed by even operation(s). These blocks summarize iterations as per convention into Extended Valuation Blocks (u,t).
3) **Structural Dynamic states** are proposed here that all Extended Valuation Blocks cluster in predictable chain families, allowing Extended Valuation Blocks to be treated as deterministic bookkeeping objects which can be factored out.This is the reduced state after quotienting by deterministic $t=1$ behavior. The induced reduced dynamics from blocks are expressed as structural steps on the state (ω,ℓ).

3. Definitions and Notation

A. The Collatz Map

We consider the Collatz map $T:\mathbb{N}\to\mathbb{N}$,

$$
T(x)=
\begin{cases}
x/2, & x\equiv 0\pmod{2},\\
3x+1, & x\equiv 1\pmod{2}.
\end{cases}
$$

All dynamics below are induced by iterates of T.

B. Extended Valuation Blocks (Bookkeeping Layer)

An **extended valuation block** is the deterministic run generated by repeatedly applying the transformation

$$
x \mapsto \frac{3x+1}{2}
$$

starting from an odd integer x, **as long as exactly one factor of 2 is removed at each step**.

Equivalently, writing

$$
x+1 = 2^{m}3^{a}\omega,\qquad \gcd(\omega,6)=1,
$$

the extended valuation block consists of the sequence of odd iterates whose valuations evolve as

$$
x_j+1 = 2^{m-j}3^{a+j}\omega,\qquad 0\le j\le m-1,
$$

until the remaining factor of 2 is exhausted. At this **tipping point**, a halving cascade of depth >1 is forced, producing a unique next odd exit.

Extended valuation blocks are deterministic and introduce no branching.
They are treated as **bookkeeping objects** and are factored out of the reduced dynamics.

C. BlockEntry Coordinates

At the bookkeeping level, an entry into an extended valuation block is represented by a pair

$$
(u,k),
$$

where:

- u is the odd seed at block entry,
- $k\in\mathbb{N}$ records the local 2-adic entry depth.

The deterministic exit of an extended valuation block defines a map

$$
E(u,k) = (u_+,k_+),
$$

called the **BlockEntry transition**. The map E summarizes the net effect of an extended valuation block followed by its forced halving cascade.

Quantities depending explicitly on (u,k) (such as $v_3(u)$) belong to this bookkeeping layer unless shown to be invariant under reduction.

D. Structural Odd Core

The **structural odd core** of a BlockEntry seed u is defined by

$$
\omega := \frac{u}{3^{v_3(u)}}.
$$

Thus $u = 3^{a}\omega$ with $a=v_3(u)$ and $\gcd(\omega,3)=1$.
Powers of 3 in u are regarded as **non-structural**, reflecting freedom in the choice of BlockEntry representative.

E. Canonical Depth Parameter

Let

$$
a := v_3(u).
$$

We define the **canonical (structural) depth**
[
\ell := k + a.
]

Interpretation:

a. (k) measures local 2-adic entry depth,
b. (a) measures the 3-adic offset of the representative,
c. (\ell) is invariant under changing representatives by powers of (3).

The definition ℓ:=k+v3(u)\ell := k + v_3(u)ℓ:=k+v3(u) is chosen so that ℓ\ellℓ is invariant under changes of BlockEntry representative that differ by powers of 3.

F. Reduced State Space

The **reduced (structural) state** is the pair
[
(\omega,\ell),
]
where:

- (\omega) is the structural odd core,
- (\ell) is the canonical depth.

The projection from BlockEntry space to reduced state space is
[
Q(u,k)
:= (\omega,\ell)
= \left(\frac{u}{3^{v_3(u)}}, k+v_3(u)\right).
]

This projection identifies all BlockEntry representatives that differ only by non-structural powers of (3).

G.Structural Steps (Reduced Dynamics)

The **reduced dynamics** are expressed as **structural steps** on the state ((\omega,\ell)).

They are induced by the BlockEntry transition via
[

$$F(\omega,\ell) := Q\big(E(u,k)\big),$$
]
where (u,k) is any BlockEntry representative satisfying $Q(u,k)=(\omega,\ell)$.

By determinism of extended valuation blocks and the definition of $\ell=k+v_3(u)$, the map F is well-defined and deterministic on the reduced state space.

H. Abstraction Convention (Lock-in Clause)

Extended valuation blocks are deterministic bookkeeping objects and are factored out.

4. Structural Step Determinism and Valuation Drift

4.1. Lemma 4.1 (Extended valuation block determinism and valuation drift)

Let (x) be an odd integer. Write
[
x+1 = 2^{m} 3^{a} \omega
]
where (m:=v_2(x+1)\ge 1), (a:=v_3(x+1)\ge 0), and (\gcd(\omega,6)=1).

Define the **(t=1) step** (one odd step followed by exactly one halving)
[
f(x);:=;\frac{3x+1}{2}.
]
For (j\ge 0), define (x_{j+1}:=f(x_j)) as long as the right-hand side is an odd integer.

Then:

1. (**Exact drift formula**) For every integer (j) with (0\le j\le m-1), the iterate (x_j) is odd and
   [
   x_j+1 ;=; 2^{m-j},3^{a+j},\omega.
   ]
   In particular, while (m-j\ge 2), the next (t=1) step is valid and keeps the iterate odd.
2. (**Tipping point**) At (j=m-1) we have
   [
   x_{m-1}+1 ;=; 2\cdot 3^{a+m-1},\omega,
   ]
   so the "tipping seed" (after removing the inherent single factor (2)) is
   [
   \frac{x_{m-1}+1}{2} ;=; 3^{a+m-1},\omega.
   ]
3. (**Cascade is forced**) The next Collatz odd step from (x_{m-1}) satisfies
   [
   3x_{m-1}+1 ;=; 2\big(3^{a+m}\omega - 1\big),
   ]
   and since (3^{a+m}\omega) is odd, ((3^{a+m}\omega-1)) is even. Hence
   [
   v_2(3x_{m-1}+1);\ge;2,
   ]
   so a halving cascade of depth at least (2) is unavoidable. Consequently there is a unique odd exit

[
x_{\mathrm{exit}} := \frac{3x_{m-1}+1}{2^{,v_2(3x_{m-1}+1)}} \quad\text{(odd)},
]
and this determines a unique next block entry.

Proof

Step 1 (key identity). For any odd (x), define (x^+=f(x)=(3x+1)/2). Then
[
x^+ + 1
= \frac{3x+1}{2} + 1
= \frac{3x+3}{2}
= \frac{3(x+1)}{2}.
\tag{★}
]

Step 2 (induction for the drift formula).
We prove by induction on (j) that for (0\le j\le m-1),
[
x_j+1 = 2^{m-j}3^{a+j}\omega.
]
Base (j=0) is exactly the factorization hypothesis (x+1=2^m3^a\omega).

Assume it holds for some (j<m). Apply (★) to (x_j):
[
x_{j+1}+1
= \frac{3(x_j+1)}{2}
= \frac{3\cdot 2^{m-j}3^{a+j}\omega}{2}
= 2^{m-(j+1)}3^{a+(j+1)}\omega.
]
This proves the formula for (j+1). Moreover, if (m-j\ge 2), then (x_j+1) is divisible by (4), so (x_j\equiv -1\pmod 4) and
[
x_{j+1}=\frac{3x_j+1}{2} = \frac{3(x_j+1)-2}{2} = 3\cdot \frac{x_j+1}{2}-1
]
is indeed odd. Thus the (t=1) iteration stays in the odd integers deterministically until (j=m-1).

Step 3 (tipping point).
Substitute (j=m-1) into the drift formula:
[
x_{m-1}+1 = 2^{m-(m-1)}3^{a+(m-1)}\omega = 2\cdot 3^{a+m-1}\omega,

]
so $(x_{m-1}+1)/2 = 3^{a+m-1}\omega$, as claimed.

Step 4 (forced cascade).
Compute the next odd step from x_{m-1} using $x_{m-1}=2\cdot 3^{a+m-1}\omega-1$:

$$
\begin{aligned}
&3x_{m-1}+1 \\
&= 3(2\cdot 3^{a+m-1}\omega-1)+1 \\
&= 2(3^{a+m}\omega - 1).
\end{aligned}
$$

Since $3^{a+m}\omega$ is odd, $3^{a+m}\omega-1$ is even, so $3x_{m-1}+1$ is divisible by 4, i.e. $v_2(3x_{m-1}+1)\ge 2$. Therefore a halving cascade of depth at least 2 occurs and yields a unique odd exit

$$
x_{\mathrm{exit}} = \frac{3x_{m-1}+1}{2^{v_2(3x_{m-1}+1)}}.
$$

Uniqueness follows because $v_2(\cdot)$ is uniquely defined.

This establishes that extended valuation blocks are deterministic and introduce no branching; hence they may be safely factored out of the reduced dynamics.∎

5. Structural Step Formula